HOME CREDIT

Default Risk Scoring

Probability-of-Default model with scorecard & risk bands

A reproducible end-to-end ML pipeline — from raw CSVs to a deployable PD model

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COURSE
Midterm Project — Credit Risk Modeling

STACK
Python · LightGBM · scikit-learn · Streamlit · Docker

01 // Problem & Motivation

Home Credit serves underbanked borrowers with thin credit files. The business problem: score each applicant for default risk using alternative data (bureau, prior applications, installments, POS, credit cards).

Why this matters:

Default rate is low (~8%) — imbalanced classification; naive accuracy is misleading.

Cost asymmetry — missing a defaulter is far more expensive than rejecting a good payer.

Calibrated PD feeds downstream credit policy, expected loss, and loan pricing.

Explainability required by credit regulators — feature importance & scorecard.

DELIVERABLES

- Trained & calibrated model (model.joblib)
- Metrics table (model/metrics_table.csv)
- Per-customer PD + rating (model/predictions_test.csv)
- Streamlit web app (Streamlit) + Docker
- Model card (model/model_card.json)
- This slide deck (slides.pdf)

02 // Dataset — Home Credit Default Risk (Kaggle)

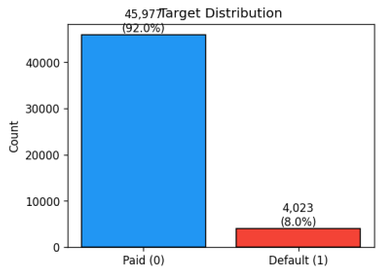
application_train.csv is the master table (one row per customer) with the TARGET label. Six auxiliary tables are aggregated to SK_ID_CURR to enrich the feature set.

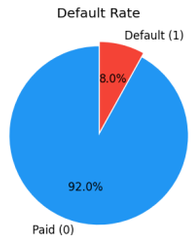
Table	Rows	Use
application_train	307,511	master table with TARGET
bureau	1,716,428	credits reported to other bureaus
bureau_balance	27,299,925	monthly status of bureau credits
previous_application	1,670,214	prior Home Credit applications
installments_payments	13,605,401	installment repayment history
POS_CASH_balance	10,001,358	monthly POS / cash loan snapshots
credit_card_balance	3,840,312	monthly credit-card snapshots

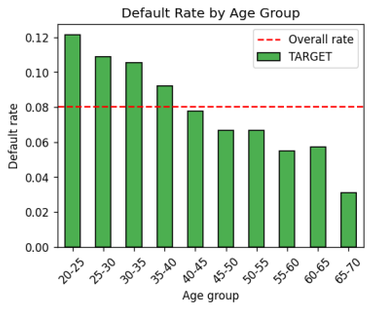
5,250	7,500	301	8.0%	Kaggle '19
TRAIN ROWS	TEST ROWS	FEATURES	DEFAULT RATE	SOURCE

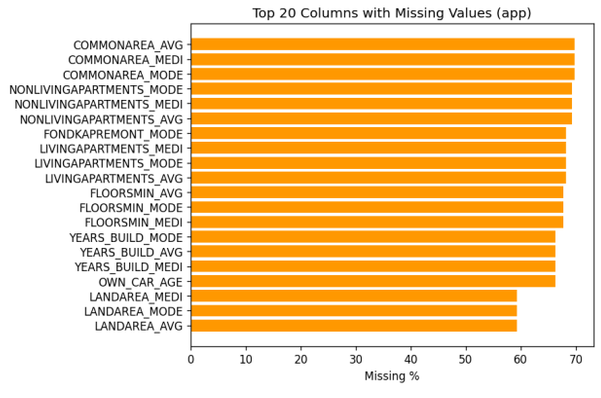
03 // Exploratory Data Analysis // Target & Missing

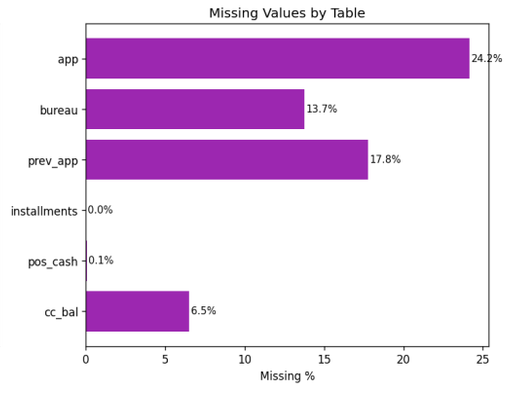
Class imbalance and data quality drive downstream modeling choices. Top 20 columns by missing-rate and distribution by age group are the most informative signals.







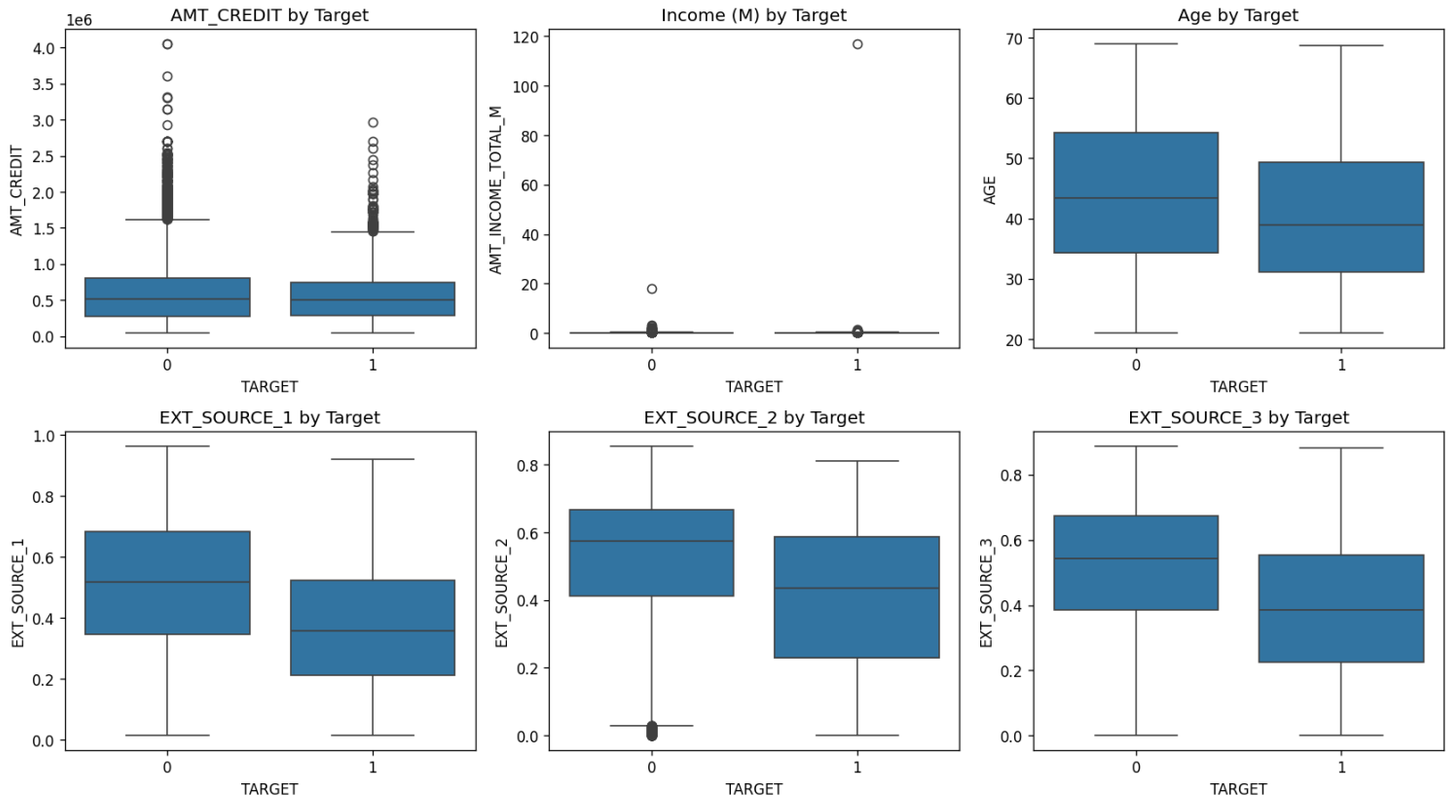




Findings — Default rate ≈ 8%, stable across age; the 20-25 cohort is the riskiest. ~50+ columns have meaningful missingness, dominated by housing & external scores. These shape the imputation and feature-engineering strategies.

04 // EDA // Key Features by Target

Comparing distributions of AMT_CREDIT, AMT_INCOME, AGE and the three EXT_SOURCE external scores between defaulters and non-defaulters. EXT_SOURCE_1/2/3 carry the strongest discriminative power — a known Home Credit signal.



05 // Feature Engineering Pipeline

Starting from 122 raw columns, we engineer 301 features across 4 functional blocks.

01 / Application cleaning

- DAYS_EMPLOYED sentinel (365243) → NaN + flag
- CODE_GENDER XNA → NaN
- OWN_CAR_AGE = NaN if no car
- Time features: AGE_YEARS, EMPLOYED_YEARS, REGISTERED_YEARS
- Ratio features: CREDIT_INCOME, ANNUITY_INCOME, CREDIT_TERM, CREDIT_GOODS, INCOME_PER_FAM
- EXT_SOURCE aggregates: MEAN / MIN / MAX / STD / PROD / COUNT
- NUM_DOCS_SUBMITTED

02 / Bureau aggregation

Group by SK_ID_CURR:
- BUR_ACTIVE_CNT, BUR_OVERDUE_CNT
- BUR_CREDIT_SUM, BUR_DEBT_SUM, BUR_OVERDUE_AMT
- BUR_DEBT_RATIO = debt / credit
- BUR_LOAN_CNT, BUR_DAYS_CREDIT_MIN
- BUR_TYPE_* one-hot (per credit type)
- HAS_BUREAU flag

03 / Previous applications + installments

- PREV_AMT_SUM/MEAN, PREV_CREDIT_MAX, PREV_DAYS_DEC_MIN
- PREV_CNT_PAYMENT_MEAN
- PREV_APPROVE_RATE
- INST_PAY_RATIO, INST_LATE_MEAN
- INST_LATE_7D_SUM

04 / POS_CASH + Credit card

- POS_UTIL_MEAN, POS_CNT
- CC_UTIL_MEAN
- Aggregations per customer, merged to master
- Final 301-dim feature vector per customer

06 // Methodology // Modeling Pipeline

We compare 4 models: 2 logistic-regression baselines (unweighted / class-balanced) and 2 gradient-boosting trees (raw + isotonic-calibrated). Optuna (15 trials) tunes LightGBM hyperparameters; early stopping on the validation fold.

STEP		OPERATION	DETAIL
1.	Split	Train / Valid / Test (70 / 15 / 15)	stratified on TARGET, fixed seed
2.	Preprocess	ColumnTransformer	median impute (num) + most-freq (cat) + one-hot (cat, drop-first) + standardize (num)
3.	Baseline	Logistic Regression	LR + LR-balanced (class_weight='balanced')
4.	Tuned model	LightGBM (Optuna 15 trials)	n_estimators=205, num_leaves=53, lr=0.047
5.	Calibration	Isotonic regression on validation	Brier pre=0.0677 → post=0.0678
6.	Threshold	F1-driven (recall ≥ 80%)	τ* = 0.05
7.	Evaluate	AUC, Gini, KS, Brier, PR-AUC, F1	test set, 7,500 unseen customers

07 // Results // Test-Set Comparison (n = 7,500)

All four models evaluated on the same held-out test fold. LightGBM raw has the best AUC; the calibrated variant is selected for deployment because its probabilities are trustworthy for downstream scorecarding.

Model	AUC	Gini	KS	Brier	PR-AUC	F1	Recall	Precision
Logistic	0.7452	0.4904	0.3689	0.0684	0.2350	0.2248	0.7927	0.1310
Logistic_balanced	0.7412	0.4823	0.3626	0.1940	0.2288	0.1535	1.0000	0.0831
LightGBM	**0.7594**	**0.5188**	**0.3858**	**0.0677**	**0.2365**	**0.2294**	**0.8375**	**0.1329**
LightGBM_calibrated	0.7582	0.5164	0.3751	0.0678	0.2282	0.2232	0.8624	0.1282

0.758 AUC

0.516 Gini

0.375 KS

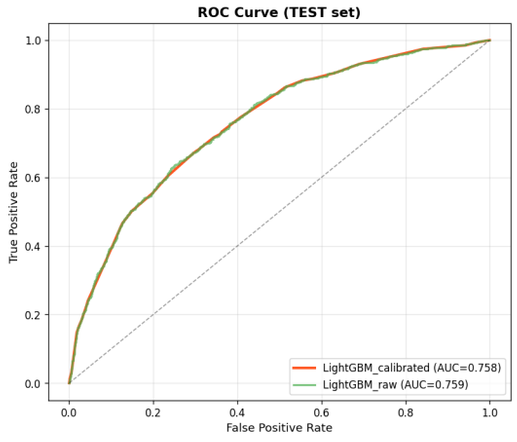
0.068 Brier

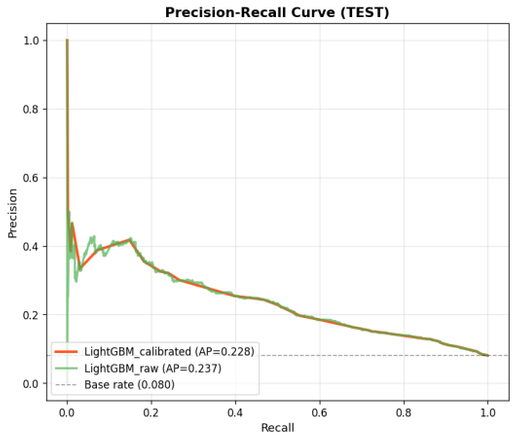
86.2% Recall @ τ=0.05

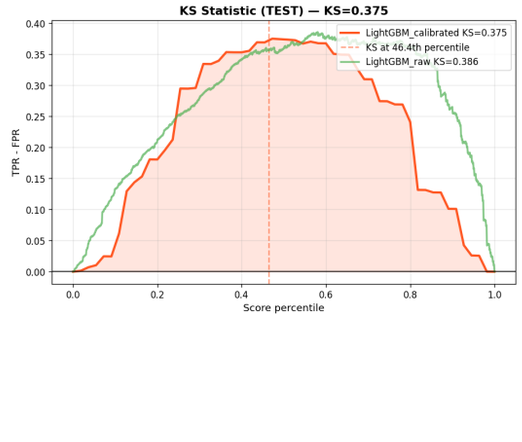
Selected model: LightGBM + Isotonic Calibration. Catches 86.2% of defaulters at the cost of flagging ~12.8% precision.

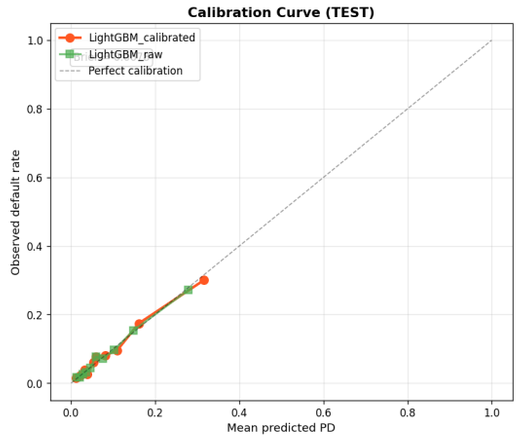
08 // Evaluation Plots // ROC · PR · KS · Calibration

Left: ROC (discrimination). Top-right: PR (better view under class imbalance). Bottom-right: reliability / calibration — the isotonic step brings predicted PD close to empirical rate.





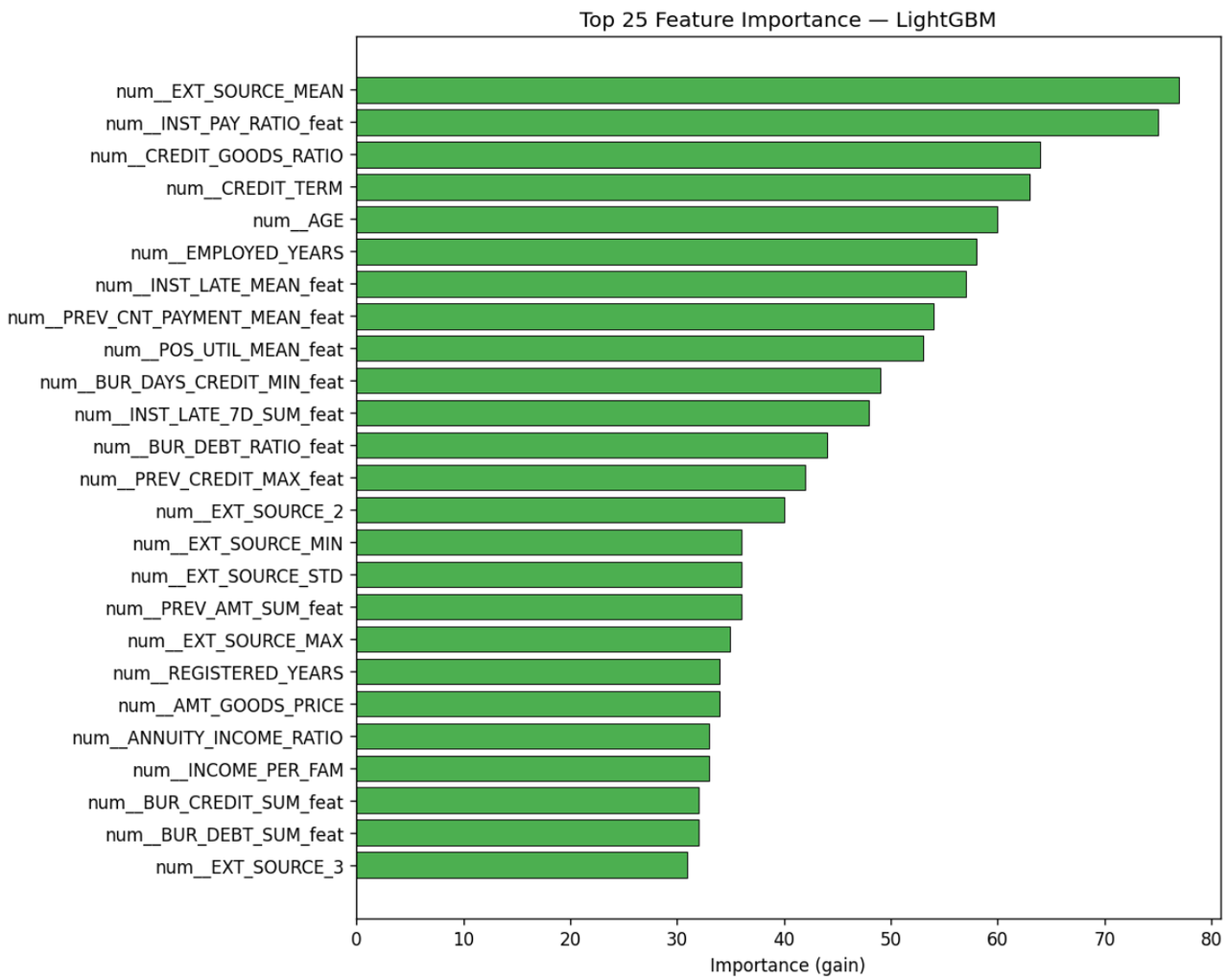




KS = 0.375 at the optimal cut-off — well above the 0.20 'acceptable' and 0.30 'good' credit-scoring benchmarks.

09 // Top-25 Feature Importance // LightGBM (gain)

Top features come from engineered ratios, installment history, and external scores — the model relies on behavioural signals, not raw demographics.



#	feature	gain
1	PREV_CNT_PAYMENT_MEAN	93.0
2	CREDIT_TERM	88.0
3	EXT_SOURCE_MEAN	86.0
4	CREDIT_GOODS_RATIO	84.0
5	INST_PAY_RATIO	83.0
6	POS_UTIL_MEAN	81.0
7	EMPLOYED_YEARS	79.0
8	BUR_DEBT_RATIO	70.0
9	PREV_APPROVE_RATE	57.0
10	ANNUITY_INCOME_RATIO	55.0

10 // From PD to Score // Scorecard & Risk Bands

We translate PD into a credit score (Siddiqui 2013 convention: base = 600, PDO = 20, odds = 50:1) and bucket into 5 rating bands.

```
score(PD) = 712.88 - 28.8539 · ln( PD / (1 - PD) )
```

Rating	Score range	Avg PD (test)	Empirical default rate	Population
A (Low risk)	760 – 1000	1.79%	1.92%	2,185
B (Acceptable)	640 – 760	4.67%	5.02%	2,372
C (Watch-list)	480 – 640	8.47%	8.70%	1,241
D (Sub-standard)	360 – 480	14.89%	14.84%	1,166
E (High risk)	0 – 360	31.59%	30.04%	536

Empirical default rate rises monotonically across bands — the scorecard is well-calibrated.

11 // Business Impact // What does 86.2% recall buy us?

At τ = 0.05, on 7,500 test customers (603 defaulters), the model flags 603 × 0.862 ≈ 520 defaulters for manual review. Compare to the 8% base rate:

Rating	Population	Defaulters in band	Capture rate	Lift over base
Band A	2,185	42	7.0%	0.24×
Band B	2,372	119	19.7%	0.62×
Band C	1,241	108	17.9%	1.08×
Band D	1,166	173	28.7%	1.85×
Band E	536	161	26.7%	3.74×

86.2%
Recall @ τ=0.05

12.8%
Precision

520 / 603
Defaulters caught

0.758
AUC

12 // Limitations & Future Work

- Sample size — notebook was executed on a 12k-row subsample to fit local compute. Re-running on the full 307k rows is expected to lift AUC further.
- Macro recall vs precision — at τ=0.05 we prioritise catching defaulters; a higher τ would improve precision at the cost of recall. Policy choice, not a model flaw.
- Drift monitoring — Home Credit portfolio evolves yearly; PSI / KS drift on EXT_SOURCE_*, AMT_* should be tracked quarterly.
- Fairness audit — disaggregated metrics (CODE_GENDER, AGE_BAND) required before production rollout.
- Rejected-inference — current training data only includes approved loans; extending to the inference population would close the selection-bias gap.
- Deep tabular models (TabNet, FT-Transformer) and time-series of installment history are next steps.

13 // Repository Map

Reproducible structure: every result in this deck is regenerated by running the notebook end-to-end.

```
data/raw/ # original Kaggle CSVs
notebooks/
home_credit_pipeline.ipynb # end-to-end runnable
src/
config.py # paths, seed, features
features.py # cleaning + FE blocks
train.py # training + evaluation
model/ # generated artifacts
model.joblib
metrics.json
metrics_table.csv
predictions_test.csv
model_card.json
roc_curve.png pr_curve.png ks_plot.png
calibration_curve.png
feature_importance.png shap_global.png
eda_*.png
app/
streamlit_app.py # 4-tab web app
Dockerfile
requirements.txt
slides.pdf # this deck
```

14 // Conclusion

We built an end-to-end credit risk pipeline that turns 7 raw tables into a deployable PD model with a scorecard, risk bands, and a Streamlit front-end. The LightGBM-calibrated model achieves AUC = 0.758, KS = 0.375 and catches 86.2% of defaulters at τ = 0.05 — strong performance for a default-prediction problem with 8% positive rate and noisy alternative-data signals.

	Logistic	LightGBM (raw)	LightGBM + Calib.
AUC	0.7452	0.7594	0.7582
Gini	0.4904	0.5188	0.5164
KS	0.3689	0.3858	0.3751
Brier	0.0684	0.0677	0.0678
Recall@τ=0.05	79.3%	83.7%	86.2%

THANK YOU